

DEEP LEARNING
Literature Review Submission
(one document per team)

Project Title	Multimodal assistive vision system for the visually impaired using spatial and vision language architectures
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1. Summary of Journal Papers

Paper 1: Edge-Based Vision-Language Assistive System for the Visually Impaired: A Quantized VLM Approach

Author(s) & Year: *A. Arystanbekov, O. Varol, and A. Yazici, 2026*

This work focuses on the problems of latency and connectivity in Vision-Language Model (VLM)-based assistive systems for the blind. The authors suggest a solution that allows the system to work offline on portable edge hardware, which lessens the requirement of cloud services. Experiments used the VizWiz and VQAv2 datasets. The proposed solution employs a quantized LLaVA-NeXT-13B model for vision-language processing, Whisper for speech processing, and PIPER for speech output. The results demonstrate that quantization reduces resource demands significantly while keeping the accuracy loss to a maximum of 2.6% according to the VizWiz dataset. However, the method still needs around 4–5 seconds per query. The key contribution of the paper is the demonstration that a VLM-based assistive tool can work on lightweight edge devices without a permanent connection to the internet. However, the relatively big latency and reliance on only one VLM hinder its potential to perform timely detection of physical objects.

Paper 2: ENVISAGE: An image captioning dataset and conformal uncertainty modeling framework to assist visually impaired individuals

Author(s) & Year: M. Çaylı, B. Kılıç, and L. Wang, 2026

The research addresses the fact that traditional image captioning datasets produce excessively general descriptions that do not have crucial spatial relationships and physical affordances necessary for navigation by the blind. The authors put forward ENVISAGE, a dedicated dataset compiled to highlight spatial layouts, object affordances, and scene context for the blind user. After that, the authors develop a spatial vision-language architecture combined with conformal uncertainty modeling that provides appropriate and contextually-dependent descriptions. The authors find that the models trained on ENVISAGE provide much more informative spatial descriptions than the foundation baselines and efficiently measure uncertainty of the predictions to prevent hallucinations. The major contribution of the work is development of a dedicated dataset with the uncertainty estimation system meant for the safe helping vision. The most important limitation of this methodology is the lack of a special real-time object detection system designed to localize physical threats and an OCR pipeline for reading words from objects.

Paper 3: AI-powered BlindSpot VisionGuide system on Raspberry Pi for enhancing independence of visually impaired users

Author(s) & Year: M. Sudha, S. Swaminathan, M. Suba, and A. Suyamburajan, 2026

The paper examines how unified and multifunctional assistive systems that can do scene description, face recognition, and reading functions offline are absent in the current market. The authors test their developed system within existing facial data sets and scene recognition benchmarks. The design exploits deep facial embeddings integrated with BLIP, an image captioning model whose operation is realized with the voice interface in Raspberry Pi 5. The experimental findings confirm that the suggested technology performs the multiple assistive tasks, ensuring that high processing speed is achieved in the conditions of minimum power consumption. The paper contributes to proving that multi-functional system can be built with lower costs. Nevertheless, the application of older versions of transformer technology such as BLIP limits the efficiency of the technology.

Paper 4: Deep learning based object detection and surrounding environment description for visually impaired people

Author(s) & Year: M. Islam et al., 2023

This study examines the inexpensive real-time location of obstacles and the basic description of surroundings for safe movement. The authors used the MS COCO collection (328,000 images) and created their own weather dataset consisting of 500 images. The proposed technology is based on the SSDLite MobileNetV2 architecture, integrated into the TensorFlow framework with its optimization by means of Gradient Particle Swarm Optimization (PSO) and executed in the form of a Raspberry Pi-based wearable head cap providing the audio output of Google Text-to-Speech application. The results demonstrate the accuracy of detection and sufficient speed of frames' transfer which allows it to use the device in real-time conditions for block avoidance. The main contribution of the paper is a lightweight and inexpensive cap for obstacles' identification. Nevertheless, the technology utilizes only object detection labels which does not permit its comprehending of the scene context, relationships between objects, reading text, etc.

Paper 5: VI-OCR: 'Visually Impaired' optical character recognition pipeline for text accessibility assessment

Author(s) & Year: Q. Gao et al., 2025

The research examines the difficulty of assessing text readability and character recognition of people with different levels of sight impairment. In terms of material, the authors used two types of recognized clinical vision charts, namely the ETDRS and the MNREAD vision tests, along with home conditions images with text. The project proposes a framework called VI-OCR that includes a visual defect module responsible for the simulation of vision impairment and a text-identification OCR module required for testing the performance of OCR systems. The findings indicate that the VI-OCR system models the reading capabilities of people suffering from vision problems under certain conditions. The core idea of the project is to provide a device that makes it possible to measure the readability of texts for the creation of appropriate devices for people with sight impairment. The VI-OCR platform is a text-analysis tool and it does not involve object recognition, scene interpretation and navigation.

2. Comparative Analysis of the Papers

S.No	Title, Author & Year	Dataset	Model Used	Advantages	Drawbacks	Your Findings
1	Edge-Based Vision-Language Assistive System. (Arystanbekov et al., 2026)	VizWiz-VQA, VQAv2	Quantized LLaVA-NeXT-13B, Whisper, PIPER	Operates offline on edge hardware without internet reliance.	High processing latency (4–5 seconds per query); lacks dedicated OCR.	Proves VLMs can run locally, but highlights the need for a faster, lightweight VLM like Florence-2.
2	ENVISAGE: An image captioning dataset. (Çaylı et al., 2026)	ENVISAGE (Custom dataset)	Spatial & VLM with Conformal Uncertainty	Captures rich spatial details and reduces model hallucinations.	Lacks direct object bounding boxes for real-time collision prevention.	Confirms VLM must provide spatial context, but requires pairing with YOLOv8 for exact obstacle localization.
3	AI-powered BlindSpot VisionGuide... (Sudha et al., 2026)	Face embeddings & Scene datasets	LIP model, Deep Face embeddings (Raspberry Pi 5)	Combines face recognition, reading, and scene captioning into one hardware unit.	BLIP struggles with complex scene text; memory limits cause execution lag.	Validates multi-task architecture, but executing on a laptop GPU avoids hardware memory bottlenecks.
4	Deep learning based object detection... (Islam et al., 2023)	MS COCO, Custom Weather dataset	SSDLite MobileNetV2 with PSO optimization	Very fast real-time frame rates and low computational cost.	Cannot provide scene context or read street signs/documents.	MobileNet provides rapid detection, but pairing YOLOv8 with a VLM gives both speed and scene understanding.
5	VI-OCR: 'Visually Impaired' optical character... (Gao et al., 2025)	ETDRS, MNREAD, Scene text images	VI-OCR pipeline (Degradation + Text Spotting)	Accurately models text reading difficulties for low-vision users.	Limited strictly to text reading; offers no navigation or spatial vision.	Demonstrates that OCR must be integrated as a dedicated module alongside spatial vision for full accessibility.

3. Problem Statement / Objective and Research Gap

3.1 Problem Statement / Objective

*Visually impaired individuals face significant challenges in navigating daily environments safely and accessing essential textual information independently. Existing assistive tools, such as basic ultrasonic canes or standalone object detectors, can identify physical obstacles but fail to interpret full scene context or extract written text from signs, labels, and documents. The objective of this project is to develop a comprehensive **Multimodal Assistive Vision System** that unifies spatial object detection, vision-language scene captioning, and optical character recognition into a single, low-latency framework. By integrating **YOLOv8** for spatial localization, **Florence-2** for compact scene understanding, **EasyOCR** for text reading, and a **Text-to-Speech (TTS)** engine for natural speech output, our system provides real-time, comprehensive audio guidance on portable hardware.*

3.2 Research Gap

The literature reveals a clear trade-off in existing assistive technologies: single-task systems execute quickly but lack context, while multimodal systems suffer from high latency and hardware bottlenecks. Works utilizing heavy Vision-Language Models (Arystanbekov et al., 2026) incur 4-to-5-second query delays and struggle with fine text, whereas pure object detection models (Islam et al., 2023) lack scene context and reading capabilities. Furthermore, multi-task systems deployed on constrained single-board hardware (Sudha et al., 2026) suffer performance degradation when handling complex transformer models alongside text pipelines. Our project fills this research gap by proposing a multi-threaded parallel architecture. By executing YOLOv8, Florence-2, and EasyOCR simultaneously on host hardware, the system eliminates processing bottlenecks, delivering low-latency, real-time spatial, textual, and contextual audio feedback to the user.

References (IEEE style)

- [1]Arystanbekov, Batyr, Huseyin Atakan Varol, and Adnan Yazici. "Edge-Based Vision-Language Assistive System for the Visually Impaired: A Quantized VLM Approach." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2026).
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- [3]Sudha, M., et al. "AI-powered BlindSpot VisionGuide system on raspberry Pi for enhancing independence of visually impaired users." *Scientific Reports* 16.1 (2026): 11316.
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